



Department of Computer Science
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Project Title: Aviation Delay Prediction System

Abstract

Predicting flight delays is very important for the aviation industry. It helps airlines run more smoothly and improves the overall experience for passengers. In recent years, machine learning has become a useful tool for this task because it can analyze large amounts of past data and identify patterns that lead to delays.

This project uses Dataset from various resources, and applies modern World technologies to summarize this data, look for patterns, study special cases, and make predictions.

Another important factor is how the data is prepared and used. Choosing the right features helps the model focus on what really affects delays. These can include departure and arrival times, airline and airport information, weather conditions, previous delays, and even holidays. When these elements are used effectively, the models can make better and more reliable predictions.

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1. Introduction

Modern aviation is considered an essential tool for connecting people across distances, ensuring safe transport of freight and conducting business across national borders and even continents. Aviation is one of the largest industries with billions of customers traveling via planes every year, connecting nations and cultures all around the globe.

Recently, the number of flights worldwide has been continuously growing despite various global challenges in the last years. The travel market has already recovered completely with current figures exceeding pre-pandemic numbers in many cases.

However, more frequent flights mean that the probability of delays increases dramatically. Besides, delays tend to cascade – a delayed plane might cause problems for further flights, thus disrupting the whole network.

The consequences of delays in flights can be serious in a highly competitive aviation industry. Delays usually imply additional expenses in form of extra use of fuel or crew management problems. At the same time, travelers can experience delays or even have their plans ruined by them.

Therefore, timely delivery of passengers to their destinations on board of the planes becomes an essential point. The airline companies compete not only based on prices but also concerning the punctuality of flights. Thus, reliable predictions of possible delays are critical.

Using machine learning algorithms, analysts and airline executives would be able to analyze massive historical databases of flights and identify specific patterns. Based on this analysis, possible delays can be detected ahead of time allowing taking precautionary measures.

2. Domain Explanation

2.1 Factors Influencing Flight Delays

According to real-life experience, flight delays are primarily influenced by exogenous and operational factors (Technical Review, 2024). One of the key causes is weather conditions that account for 52.43% of all flight delays because of the associated danger coming from storms, fog, and precipitation. Secondly, air traffic is another key factor causing 35.47% of delays. It reflects the overcrowding of runways and the atmosphere during peak periods.

Other causes of delays include runway closures (5.78%), other reasons (5.47%), and aircraft equipment malfunctions (0.86%). Moreover, operational factors are also important, such as the availability of pilots and crews and delay cascading when one flight becomes delayed, affecting the following flights(Review Panel, 2023) .

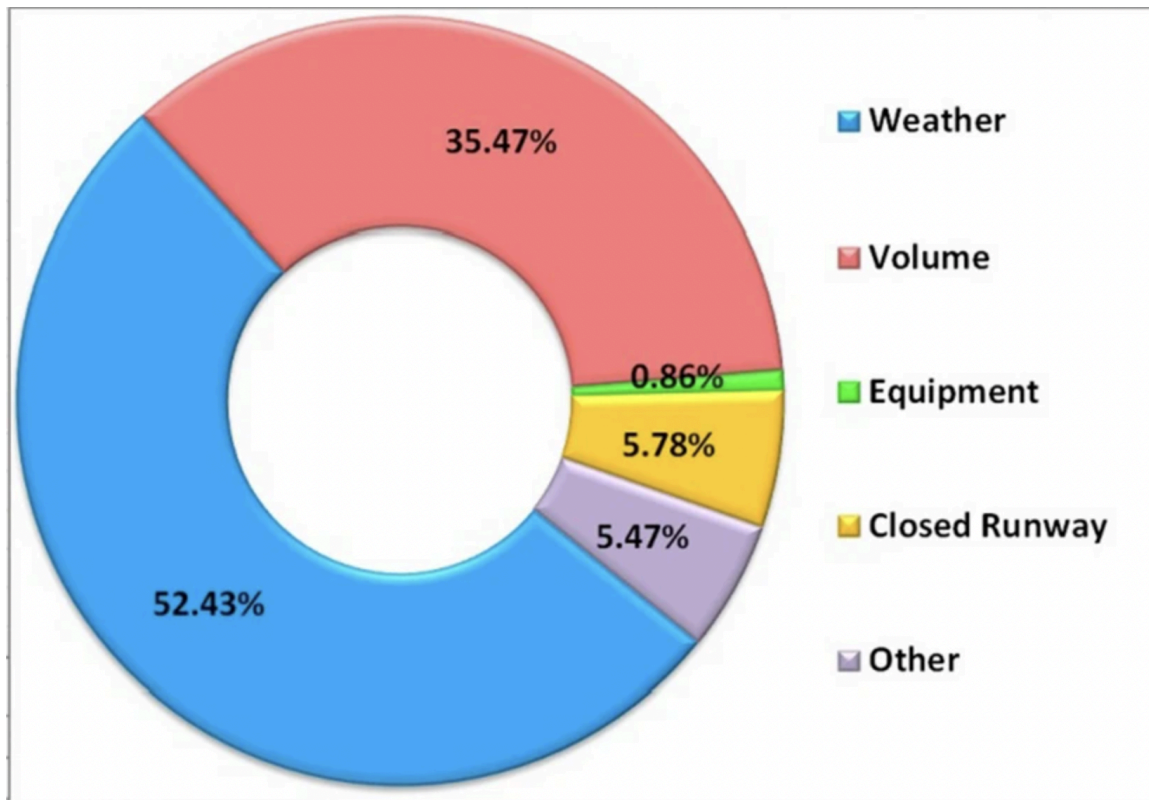


Figure 1: Distribution of primary causes for flight delays.

2.2 Economic and Environmental Impact

Beyond operational disruptions, flight delays result in substantial financial expenses due to increased fuel consumption and related costs linked to passengers. For instance, delays cost the aviation industry billions of dollars annually (e.g., an estimated \$33 billion in 2019) and lead to the waste of billions of liters of jet fuel while aircraft idle. This massive economic and environmental burden underscores the necessity for our highly accurate predictive models(Wang et al., 2024) .

3. Algorithm Background

3.1 Machine Learning Methodology

The Aviation Delay Prediction System project is framed as a Supervised Machine Learning problem. To ensure a comprehensive analysis, our team will implement a dual-modeling strategy(Erdem & Bilgiç, 2023) :

- Classification: To predict a binary outcome (e.g., distinguishing between "Delayed" and "On-Time" flights) based on the standard 15-minute threshold.
- Regression: To estimate the continuous duration of delay minutes, which provides actionable data for airport resource allocation.

3.2 Model Selection & Justification

Based on recent academic benchmarks (2021–2026), we have selected the following models for implementation:

- Extreme Gradient Boosting (XGBoost): This is our primary candidate for high performance. Recent 2026 studies demonstrate that XGBoost significantly outperforms traditional Deep Learning models when processing structured aviation data, offering better accuracy and faster training times(Comparative Analysis, 2026) .
- Random Forest (RF): We will utilize this ensemble method to handle the non-linear relationships within our 15+ features (such as the interaction between weather and airport congestion). RF is particularly effective in reducing overfitting, which is common in complex multi-country datasets.
- Baseline Models: We will implement Linear and Logistic Regression to serve as performance baselines. This allows us to quantify the exact improvement offered by more advanced ensemble techniques.

3.3. Feature Engineering and Implementation

To enhance model precision, we will prioritize the Previous Flight Delay (PFD) feature. Research indicates that the delay status of an aircraft's preceding journey is often the strongest predictor of its next arrival time. The implementation will be conducted using Python, specifically leveraging the Pandas library for data manipulation and Scikit-learn for model training and evaluation, ensuring strict adherence to course technical rules.

3.4 Evaluation Metrics

To assess model success, we will use metrics that account for data imbalance:

Accuracy & F1-Score: For the classification task to ensure balanced performance between precision and recall.

Root Mean Square Error (RMSE): For regression to quantify the deviation between predicted and actual delay durations.

4. References

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